DCR2a: Class-Based Scoring Wins at 1904 by One Position, Then Fails the Placebo

Human director: Jawaun Brown **Producing agent:** Claude Code, directed **Program:** Date-Cut Retrodiction — DCR2a (empirical companion to DR5) **Status:** Overall **NO_GO** on the preregistered gate suite. N3 GO (class-scoring surfaces T1 above the proposition-blind baseline at 1904), N4 **NO_GO** (class-scoring also surfaces T1 at rank 2 at the 1880 deep placebo). This outcome is the DR5 theorem operating: class-aware nomination adds a grouping function g, and when g produces low-cardinality classes with borderline members, the projection-vs-realisation ambiguity is inherited from the ranker to g. **Date:** 2026-07-27

Abstract

DCR1f established that T1 (absolute simultaneity) is not a discrete proposition to look for — it is a class of six or more non-equivalent surface realisations across the pre-1905 corpus, and no proposition-ranking matcher can catch them all at once. DR5 (parallel theorem paper) formalises this: a proposition-ranking nominator cannot distinguish a commitment D from any specific realisation r_i, and any escape requires importing an external grouping function g.

DCR2a is DR5's empirical companion. It applies class-based scoring to the DCR1e presupposition-extracted consensus, using target_v4 as g, and reports the load-bearing comparison: does class-based ranking surface T1 in a way proposition-blind ranking does not?

The answer is a qualified yes and a definitive no. Yes, class-based scoring puts the T1 class at rank 3 at the 1904 target cut, one position above where the best T1 realisation lands proposition-blind (rank 4). That is a real, if narrow, improvement.

No, the improvement does not survive the placebo. Class-based scoring under all three preregistered aggregation rules (cardinality, coverage, spread) also puts T1 at rank 2 at the 1880 deep placebo cut, where target_v4 fires exactly once (Maxwell 1865's instant across whole medium). Any single-hit class dominates cardinality in a corpus of small classes; the placebo firing that DCR1f identified as either extractor projection (B1) or invalid-placebo (B2) is preserved, not resolved, by class aggregation.

The theorem predicts this. Class-aware nomination adds g (which resolves the D-vs-r_i ambiguity when g is sound and complete for D), but it inherits the correctness question about g from the ranker level. On this run, g = target_v4 is empirically not sound at the 1880 cut — either because Maxwell's field-theoretic "*instant at which the whole medium…*" is a legitimate T1 realisation and the placebo was never valid for T1, or because target_v4 is over-firing at the placebo. The two readings remain formally indistinguishable, as DR5 §5 shows they must in any protocol where D admits multiple non-equivalent surface forms.

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1. What was preregistered

DCR2A_PREREGISTRATION.md (2026-07-27, before nominate_by_class.py was drafted and before run_dcr2a.py was executed) fixed:

N3 T1 class rank strictly better than proposition-blind's best T1 realisation rank, **N4 T1 class rank ≥ 3 at 1880 under every rule**, N5 comparison table produced.

"winner" selection: cardinality, coverage, spread.

scoring fails placebo; aggregation rule must require multi-document coverage."

redesign after seeing the result.

- Five gates: N1 class-assignment complete, N2 T1 class size ≥ 5 at 1904,

- Three aggregation rules to be reported in parallel with no post-hoc
- A decision table binding N3 GO + N4 NO_GO to the reading "class
- A single-shot commitment: no replay knobs, no aggregation-rule

Nothing here was tuned after results came in.

2. Results

Class sizes at the target cut (1904 electrodynamics scope):

class	size	documents	proposition-blind best rank
unclassified	235	15	1
T2 (privileged frame)	11	8	13
T1 (absolute simultaneity)	7	5	4
T3 (local time artifice)	2	2	38

Total propositions at 1904: 254. Total documents: 15.

Class ranks under each aggregation rule:

rule	T1 rank at 1904	T1 rank at 1880
cardinality	3	2
coverage	3	2
spread	3	2

Proposition-blind best rank at 1904 (score = statement word count, a deliberately un-tuned baseline): T1's best realisation at position 4.

3. Gate decisions

gate		
N1 class assignment complete	GO	235 / 254 unclassified (< 95%)
N2 T1 class non-empty at 1904	GO	7 members, threshold was 5
N3 T1 class rank beats proposition-blind	**GO**	rank 3 vs proposition-blind rank 4
N4 T1 class silent at 1880 placebo	**NO_GO**	T1 at rank 2 under every rule
N5 comparison table produced	GO	see §2

Overall NO_GO. Licensed reading (from the runner):

> *class_scoring_fails_placebo: T1 outranks other classes at 1880 under*
>
> *some rule, meaning cardinality-1 hits inflate the class rank.*
>
> *Aggregation rule must require multi-document coverage.*

4. Reading in light of the DR5 theorem

DR5 §4 predicted this shape exactly. The theorem says that class-aware nomination requires a grouping function g that partitions P into classes containing $\mathrm{realisations}(D, C)$. The ranking question — "is D ranked highly?" — becomes well-posed once g is supplied. But the theorem is silent on *whether g is sound and complete*.

DCR2a chose g = target_v4 (the DCR1f matcher successor). target_v4 fires on 7 propositions at the 1904 cut, including all five DCR1e T1-content propositions plus Lorentz's corresponding_instants and Poincaré 1898's same_causes_same_time. It also fires on Maxwell 1865's instant across whole medium at 1880.

The 1904 hits are largely sound: five have been individually adjudicated in DCR1e and DCR1f. The 1880 hit is the boundary case DCR1f §3 articulated:

presuppose T1 in the sense that Einstein deleted; the extractor produced the sentence, the matcher fired on it, but the underlying content is a mathematical formalism of field theory that does not constitute a specific claim about absolute simultaneity across observers.

1865 does presuppose T1: "an instant at which the whole medium…" quantifies over a spatially extended field at a single time coordinate, and this construction is a legitimate field-theoretic realisation of T1.

- **B1** — target_v4 is projecting. Maxwell 1865 does not really
- **B2** — target_v4 is right and the placebo was never valid. Maxwell

Class-based scoring does not distinguish B1 from B2. It cannot, by DR5: the difference is about whether g is sound at Maxwell 1865, and g's soundness is not a function of the ranker's scores. DCR2a inherits the DCR1f ambiguity; that inheritance is a *feature* of the theorem, not a failure of DCR2a.

5. What DCR2a establishes

scoring surfaces T1 at rank 3 where proposition-blind scoring gives T1's best realisation rank 4. On a corpus of 254 propositions in a target cut, one-position improvement in the class ranking is measurable but small. The two questions this run separates — *does class-scoring escape the DCR1f wall in general?* vs *how much of the wall does it escape on this corpus?* — receive different answers.

small classes.* *Every T1 class at 1904 had at most 11 members (T2), and at 1880 T1 had 1 member; cardinality-1 hits inflate class rank uniformly across rules.* coverage *and* spread *did not fix the problem because in this corpus |members| and |documents-per-class| are almost proportional. A rule that* requires* multi-document coverage (e.g., ignore classes with fewer than 2 documents) was considered but declined post-hoc; the paper honors the preregistration and reports the failure rather than redesigning.

class level.* *This is the DR5 §5 point instantiated: class-based scoring does not delete the ambiguity, it moves it from* "did N find the right proposition?" *to* "is g sound at the placebo cut?"* The move is progress in one sense (the D-vs-r_i collision is gone) and no progress in another (a new correctness question replaces the old one).

- **The one-position gain is real, and it is not much.** Class-based
- **Cardinality is not the right aggregation rule for corpora with
- **The projection/placebo-invalidity ambiguity persists at the

6. Two subsidiary observations

Unclassified is #1 by every rule. With 235 of 254 propositions unclassified, unclassified dominates cardinality-based rankings by a factor of 20-plus. This is not a problem for DCR2a — the load-bearing comparison is between named facet classes and proposition-blind ranking — but it is a real observation: target_v4 as a facet system leaves the overwhelming bulk of the extracted consensus unassigned. Class-aware nomination as this run implements it is an interior refinement, not a full partition of the corpus into meaningful classes. A DCR2b or DCR3 would need a facet vocabulary broader than {T1, T2, T3}.

T3 at 1904 is not helped by class scoring. T3 has 2 members and gets rank 4 by cardinality (below T1's rank 3). Its best proposition-blind rank is 38. So class-scoring dramatically improves T3's rank (from 38 proposition-blind to 4 class-based), which is a much larger fractional gain than T1's. But T3 has only two propositions; the gain is fragile. DCR2a's N3 was specifically about T1, so this observation is offered as a note rather than a claim.

7. What DCR2a does not license

DCR2a tested only the class-vs-proposition ranking comparison at a scoring level, not the deletion-repair pipeline that DR1–DR4 built.

DCR2a is one corpus, one grouping function, three aggregation rules. DR5 requires only the structural direction of the escape (class primitive is a strict generalisation of proposition primitive) not its empirical adequacy on any specific corpus.

DCR1e NO_GO, DCR1f NO_GO. DCR2a adds one measurement (the rank-3 class placement of T1 at 1904 under this specific g) and reports that it does not survive the placebo.

- **DCR2 (full deletion-repair nomination) on the enriched consensus.**
- **A general claim that class-based scoring escapes DR5's wall.**
- **A revised verdict on DCR1c/e/f.** Prior verdicts remain: DCR1c GO,

8. Next work

Two directions, both preregistered as future runs and not attempted here:

scoring with an aggregation rule that treats singleton classes as ties for last. Rerun on DCR1e consensus. If N4 (placebo) now GOes, the DR5 corollary about g's soundness at low cardinalities receives empirical support that was not preregistered here.

extractor as g itself, rather than as an intermediate step feeding target_v4. If the extractor's own class-labelling of propositions produces a cleaner T1 class than target_v4 does, the theorem's "implicit g from extraction" open question receives an empirical vote.

- **DCR2b — multi-document coverage requirement.** Reimplement class
- **DCR2c — different grouping function.** Use a presupposition-inferring

Neither replaces DR5's theorem-shape claim. Both would be honest follow-ups to DCR2a's specific result.

Appendix: reproduction

```
uv run --no-sync python -m experiments.date_cut_retrodiction.run_dcr2a
```

Reads the DCR1e presupposition-extraction consensus, assigns classes via target_v4, scores under all three aggregation rules, ranks, computes proposition-blind baseline (score = statement word count), reports the comparison. Local CPU, seconds. Writes results/dcr2a_verdict.json.

Preregistration digest (SHA-256 of DCR2A_PREREGISTRATION.md**):** 76d95494ca8968814a062f0b6bd9ceb287fcb3c135fe8647186b387a87a96717.